EAST WEST UNIVERSITY

Department of Computer Science and Engineering

LABORATORY REPORT

CHE109 — Engineering Chemistry

Experiment No.	03
Experiment Title	Standardization of $KMnO_4$ Solution with Standard $H_2C_2O_4$ Solution
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Submitted to: Course Instructor
Department of Mathematical and Physical Sciences

Experiment 03: Standardization of $KMnO_4$ Solution with Standard $H_2C_2O_4$ Solution

Objective

To determine the concentration of supplied potassium permanganate ($KMnO_4$) by titrating it against standard 0.05 M oxalic acid ($H_2C_2O_4$) in acidic medium.

Introduction

A redox reaction involves electron transfer between chemical species. In this experiment, permanganate ions are reduced to Mn^{2+} in acidic medium while oxalate ions are oxidized to carbon dioxide. The oxalic acid solution is warmed because the reaction is slow at room temperature.

Overall reaction:
$2KMnO_4 + 5H_2C_2O_4 + 3H_2SO_4 \rightarrow K_2SO_4 + 2MnSO_4 + 10CO_2 + 8H_2O$
Net ionic equation:
$2MnO_4^- + 5C_2O_4^{2-} + 16H^+ \rightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$

Chemicals and Apparatus

Chemicals: $KMnO_4$, $H_2C_2O_4$, H_2SO_4.
Apparatus: Conical flask, volumetric flask, burette, dropper, funnel, pipette, pipette filler, etc.

Procedure (copied word by word from the images)

a) Preparation of 250 mL 0.05 M $H_2C_2O_4$ solution
Weigh out of 1.575 gm of $H_2C_2O_4$ by following this formula, w=SMV/1000 (where, equivalent weight of $H_2C_2O_4$ is 126.000 g/mol) in 250 mL volumetric flask and add small- amount of distilled water. Shake the flask to dissolve the solute. Finally add more water up to 250 mL mark.

b) Standardization of $KMnO_4$ solution
1. Fill out the burette with $KMnO_4$ solution.
2. Take 10 mL of 0.05 M standard $H_2C_2O_4$ solution in a conical flask.
3. Add 5mL 1 M H_2SO_4 to the $H_2C_2O_4$ sample.
4. Heat the acidified $H_2C_2O_4$ solution in water bath for 5 minutes to about 60-75 °C.
5. Record the initial burette reading. Titrate the hot solution with the $KMnO_4$ solution until the appearance of a faint pink color. Record the final burette reading and calculate the volume of $KMnO_4$ used in the titration.
6. Repeat the titration two more times.

Experimental Data

Standard $H_2C_2O_4$ concentration = 0.050 M; aliquot volume = 10.00 mL.

Observation	IBR (mL)	FBR (mL)	Difference (mL)
1	0.4	11.0	10.6
2	11.0	21.1	10.1
3	21.1	30.5	9.4
Average	—	—	10.03

Calculations

Average titre: (10.6 + 10.1 + 9.4) / 3 = 10.03 mL.

Concentration of $KMnO_4$: 2 moles $KMnO_4$ react with 5 moles $H_2C_2O_4$.
$M(KMnO_4) = (2/5 \times 0.050 \times 10.00) / 10.03 = 0.01994$ M.

Percentage Error

% error = |theoretical − experimental| / theoretical × 100
= |0.02000 − 0.01994| / 0.02000 × 100 = 0.30% approximately.

Result

The concentration of the supplied $KMnO_4$ solution was determined to be approximately 0.01994 M, with an average titre of 10.03 mL.

Discussion / Error Analysis

The titres were 10.6, 10.1 and 9.4 mL, showing some variation between trials. The calculated concentration is close to the nominal 0.0200 M value. Possible errors include overshooting the faint-pink endpoint, inaccurate burette readings, unconditioned glassware, transfer loss or insufficient warming. The reaction is slow when cool, so maintaining 60–75 °C helps. Accuracy can be improved by adding $KMnO_4$ dropwise near the endpoint, swirling continuously, reading at eye level and repeating titrations until readings agree closely.

Answers to Practice Questions

1. What is redox titration?

A volumetric method based on an oxidation–reduction reaction involving electron transfer between titrant and analyte.

2. Why is no indicator used in this titration?

$KMnO_4$ acts as a self-indicator. A slight excess gives a faint persistent pink colour at the endpoint.

3. How does $KMnO_4$ act as a self-indicator?

Purple MnO_4^- is reduced to nearly colourless Mn^{2+} in acidic solution. At the endpoint, excess $KMnO_4$ produces a faint pink colour.

4. What happens if the titration is conducted at room temperature?

The reaction is slow, making the endpoint harder to judge and possibly causing inaccurate results. Warming to 60–75 °C increases the reaction rate.

5. Discuss the precautions of the above experiment.

Wear a lab coat and follow instructions. Use clean glassware and rinse the burette with $KMnO_4$. Remove air bubbles and read the meniscus at eye level. Measure solutions accurately. Warm the acidified oxalic acid in a water bath; do not heat directly over a flame. Swirl and add $KMnO_4$ dropwise near the endpoint. Stop at the first faint persistent pink colour, record readings carefully and clean equipment after use.

Submission reminder

Attach the instructor-signed manual page. It was not included in the provided images. Fill in the submission date before submitting.